

Surgical Knowledge Swapping in Large Language Models: Causal Localization and Precision Model Editing

Abhishek Soundalgekar Aryan Sawant Prathiksha Nijamkari Rhesha Vinod Siddhant Salian

University of Southern California

{soundalg, aryanavi, ravibabu, rvinod, ssaliam}@usc.edu

1. Abstract

Large Language Models (LLMs) encode factual knowledge in parameters learned during pre-training; when facts change, full retraining is prohibitively expensive and risks catastrophic forgetting. This paper presents our work on **surgical knowledge swapping**, targeted modification of specific factual associations in pretrained transformers, comparing three editing methods on **Qwen2.5-1.5B-Instruct** [4] using **100 CounterFact** edits [1]: ROME [1], MEMIT [2], and LoRA [3].

Causal tracing identified **Layer 10** as the critical factual memory layer, achieving **213.0%** probability recovery, confirming that factual recall in transformers is dominated by a single MLP layer. Building on this finding, ROME applies a rank-one weight update at Layer 10 and achieved near-perfect edit success (**ESR = 0.98**) with strong paraphrase generalization (**PG = 0.73**). MEMIT distributes the update across five layers (8–12), achieving perfect edit success (**ESR = 1.00**) with superior locality preservation (**LL = 1.00**) and low sequential interference (**IR = 0.04**). In contrast, LoRA adapter fine-tuning, despite training only 0.047% of parameters, fails as a surgical editor (**ESR = 0.40**, **LL = 1196.66**), confirming that parameter efficiency does not imply edit precision.

Our results demonstrate that closed-form, localized weight updates are fundamentally more suitable for surgical knowledge editing than gradient-based fine-tuning, and we provide a unified five-metric evaluation framework (ESR, PG, LL, CS, IR) for systematic comparison of editing methods.

2. Introduction

LLMs encode vast factual knowledge within their parameters during pre-training [9]. As the world evolves, this knowledge becomes stale, but retraining is prohibitively expensive and naive fine-tuning risks catastrophic forgetting [10].

Surgical knowledge editing offers a targeted alternative: modify specific factual associations in model weights while leaving all other capabilities intact. This requires understanding exactly *where* individual facts are stored, motivating the use of **causal tracing** (activation patching) to identify the critical computation nodes.

Prior mechanistic work has shown that transformer MLP layers function as key-value memories [5], where individual neurons encode specific factual associations [6]. Building on this, ROME [1] applies rank-one weight updates to precisely rewrite stored associations, while MEMIT [2] extends this to batch and sequential settings by distributing updates across multiple layers. We also test LoRA [3] to determine whether parameter efficiency translates to surgical precision.

Our contributions:

1. **Causal tracing** of Qwen2.5-1.5B-Instruct: Layer 10 identified as the critical factual storage layer (213.0% recovery).
2. **ROME** single-edit implementation with quantization-compatible hook injection (ESR = 0.98, PG = 0.73).
3. **MEMIT** batch and sequential editing with low interference (IR = 0.04) and superior locality (LL = 1.00).
4. **LoRA** baseline showing adapter fine-tuning fails surgically (ESR = 0.40, LL = 1196.66).
5. A **unified five-metric framework** (ESR, PG, LL, CS, IR) evaluated across 100 CounterFact edits.

3. Related Work

Factual Storage in Transformers.. Geva et al. [5] showed that feed-forward MLP layers act as key-value memories. Dai et al. [6] localized this to individual “knowledge neurons,” providing the mechanistic basis for targeted editing.

Knowledge Editing.. ROME [1] combines causal tracing with a closed-form rank-one update to rewrite factual associations. MEMIT [2] extends this to batch and sequential editing by distributing updates across multiple MLP layers, reducing per-layer perturbation. MEND [7] trains a hypernetwork to predict gradient-based updates; KnowledgeEditor [8] uses constrained fine-tuning.

Parameter-Efficient Fine-Tuning.. LoRA [3] augments frozen model weights with trainable low-rank matrices, enabling efficient task adaptation. Its suitability for surgical fact editing, where locality is paramount, has not been systematically quantified prior to our work.

Evaluation.. CounterFact [1] provides 21,919 counterfactual edits with paraphrase prompts and locality probes. We extend its standard metrics (ESR, PG, LL) with Consistency Score (CS) and Interference Rate (IR).

4. Problem Description

4.1. Task Formulation

Given model $\mathcal{M}$ with parameters θ , factual triple (s, r, o) , and desired new object o^* , produce $\mathcal{M}'$ with θ' satisfying:

1. **Efficacy:** $\mathcal{M}'$ generates o^* given prompt (s, r) .
2. **Generalization:** Edit transfers to paraphrased (s, r) prompts.
3. **Locality:** Unrelated factual associations remain unchanged.
4. **Non-interference:** Sequential edits do not degrade each other.

4.2. Model

We use **Qwen2.5-1.5B-Instruct** [4], a 28-layer decoder-only transformer with 1,544M parameters (889M in 4-bit mode) and grouped query attention, loaded in **4-bit NF4 quantization** via bitsandbytes on an NVIDIA T4 GPU. Because 4-bit quantized weights cannot be modified in-place, all edits are applied as **registered PyTorch forward hooks**, mathematically equivalent to direct weight modification.

4.3. Dataset

We evaluate on **100 diverse CounterFact facts** [1] covering a wide range of factual categories including language, nationality, religion, musical instrument, country affiliation, sports position, and city association. Each fact includes 2 paraphrase prompts (for PG) and 1 unrelated probe (for LL). Random seed 42

is fixed across all libraries; all prompt–output pairs are logged to `run_log.jsonl`.

4.4. Evaluation Metrics

- **ESR:** Does the model generate the new target?
- **PG:** Fraction of paraphrase prompts for which the edit holds.
- **LL:** $|P_{\text{after}} - P_{\text{base}}| / P_{\text{base}}$ (disruption to an unrelated fact; lower is better).
- **CS:** $P(\text{new}) > P(\text{old})$ measured immediately after each method’s edit is applied (i.e., ESR soft, per method).
- **IR:** Fraction of prior edits broken by each new sequential edit (MEMIT sequential experiment only).

5. Methods

5.1. Task 1: Environment & Data Setup

Qwen2.5-1.5B-Instruct is initialized in 4-bit NF4 quantization on CUDA, falling back to float16 on Apple MPS and float32 on CPU. A reusable `generate_text()` function performs greedy decoding. Seeds are fixed (42) across PyTorch, NumPy, and Python, and every prompt–output pair is logged for reproducibility. Baseline unrelated-fact probabilities are recorded before any editing.

5.2. Task 2: Causal Tracing

We apply the three-run activation patching procedure [1]:

- (1) **Clean run:** Standard forward pass; record $P(\text{target}|\text{prompt})$.
- (2) **Corrupted run:** Inject Gaussian noise ($\sigma=0.1$) into every layer output at subject token positions; $P(\text{target})$ drops significantly.
- (3) **Patched runs:** For each layer ℓ , restore the clean hidden state at the subject’s final token and measure recovery:

$$\Delta P_\ell = P_{\text{patched}}(\text{target}) - P_{\text{corrupted}}(\text{target}) \quad (1)$$

Layer 10 achieves maximum recovery of **213.0%** of the clean baseline (exceeding 100% due to compensatory effects). All other layers show substantially lower recovery, confirming that factual recall is dominated by a single critical MLP layer [1].

5.3. Task 3: ROME

ROME [1] treats the Layer-10 MLP as a key-value memory [5]. For each edit $(s \rightarrow o^*)$:

Key extraction: Capture pre-MLP activation $\mathbf{k} \in \mathbb{R}^d$ at the subject’s last-token position.

Value optimization: Run 25 Adam steps ($\text{lr}=0.4$) to find $\mathbf{v}^*$ minimizing cross-entropy loss for target o^* .

Rank-one update:

$$\Delta W = (\mathbf{v}^* - \mathbf{v}_0) \otimes \frac{\mathbf{k}}{\mathbf{k}^\top \mathbf{k}} \quad (2)$$

Implementation note: The original ROME formulation uses a corpus-estimated covariance matrix $C = \mathbb{E}[\mathbf{k}\mathbf{k}^\top]$. Our implementation uses $C = I$ (identity) as a placeholder; computing C from $\sim 1,000$ diverse sentences is deferred to future work.

Applied as a forward hook on `layers[10].mlp.down_proj`; each edit is evaluated independently and the hook is removed after evaluation, fully restoring the model between experiments.

5.4. Task 4: MEMIT

MEMIT [2] distributes the update across $L=5$ layers (8–12), centered on the causal tracing best layer, each receiving scale $s = 1/L = 0.2$:

$$\Delta W_\ell = \frac{1}{L} (\mathbf{v}_\ell^* - \mathbf{v}_{0,\ell}) \otimes \hat{\mathbf{k}}_\ell, \quad \ell \in \{8, \dots, 12\} \quad (3)$$

Spreading the update reduces per-layer perturbation, improving locality. Key vectors are projected using a regularised covariance matrix $C = K K^\top + \lambda I$ (with adaptive λ), solved on CPU to avoid GPU OOM; single-fact updates use the Sherman–Morrison formula. Two experiments: (a) **batch edit** (all 100 facts at once), and (b) **sequential edit** (facts one at a time; all prior edits verified after each addition).

5.5. Task 5: LoRA Baseline

LoRA [3] attaches trainable low-rank adapter matrices to the frozen `down_proj` layers at layers 13–17:

$$h = W_{\text{frozen}} \mathbf{x} + (\mathbf{x} A^\top B^\top) \cdot \frac{\alpha}{r} \quad (4)$$

with $r=8$, $\alpha=16$, training only **419,840 parameters** (0.047% of 889M). All 100 facts are fine-tuned jointly for 4 epochs with AdamW ($\text{lr} = 5 \times 10^{-4}$), prompt-masked cross-entropy loss. Training loss: $6.47 \rightarrow 3.40$ over 4 epochs.

6. Experimental Results

6.1. Causal Tracing

Patching across all 28 layers confirms **Layer 10** as the dominant factual memory layer (213.0% recovery). All other layers show substantially lower recovery, consistent with prior observations in GPT-family models [1].

6.2. ROME Single-Edit Results

ROME is evaluated on all 100 CounterFact edits. Each edit is applied independently and the model is fully restored between experiments.

Table 1: ROME aggregate results across 100 CounterFact edits. Model restored between each experiment.

Metric	Mean	Std	Min	Max
ESR (hard)	0.980	0.141	0.000	1.000
ESR (soft)	1.000	0.000	1.000	1.000
PG	0.730	0.358	0.000	1.000
LL ratio	1.500	5.081	0.009	51.770

ROME achieves **ESR = 0.98** (98 of 100 edits successful) and perfect soft ESR of 1.00 across all edits, with $P(\text{new}) > P(\text{old})$ in every case. PG averages 0.73, strong paraphrase transfer in the majority of edits. The mean LL is 1.500, but this is driven by a small number of outliers (max LL = 51.77), motivating the multi-layer design of MEMIT.

6.3. MEMIT Results

Table 2: MEMIT batch-edit results: 100 facts simultaneously across layers 8–12.

Metric	MEMIT	ROME
ESR	1.000	0.980
PG	0.555	0.730
LL ratio	1.001	1.500

MEMIT achieves **perfect ESR = 1.00** (surpassing ROME’s 0.98) while reducing mean LL from 1.500 to **1.001**, a 33% improvement in locality preservation. PG drops from 0.73 to 0.555, exposing a generalization–locality trade-off: spreading the edit across 5 layers reduces per-layer signal strength, limiting paraphrase transfer.

Sequential Interference.. The mean interference rate across all sequential additions is **IR = 0.041**, indicating that each new edit breaks only $\sim 4\%$ of previously applied edits on average. This demonstrates that MEMIT’s multi-layer distribution strategy maintains high stability even across 100 sequential edits.

6.4. LoRA Baseline Results

LoRA is evaluated on the same 100 CounterFact edits, fine-tuned jointly for 4 epochs with AdamW ($\text{lr} = 5 \times 10^{-4}$).

Table 3: LoRA baseline results ($r=8$, $\alpha=16$, 4 epochs, 100 facts).

Metric	LoRA	ROME
ESR	0.400	0.980
PG	0.140	0.730
LL ratio	1196.66	1.500

LoRA succeeds on only 40 of 100 facts (ESR=0.40). Mean LL reaches **1196.66**, orders of magnitude worse than ROME (1.50) or MEMIT (1.00). Despite training only 0.047% of parameters, LoRA’s gradient steps reshape representations globally across all token contexts, causing widespread disruption to unrelated knowledge pathways.

6.5. Unified Comparison

Table 4: Cross-method comparison on all five metrics (100 CounterFact edits). CS = $P(\text{new}) > P(\text{old})$ measured immediately post-edit per method; “—” denotes metric not applicable.

Method	ESR	PG	LL	CS	IR
ROME	0.98	0.73	1.50	1.00	—
MEMIT	1.00	0.56	1.00	1.00	0.04
LoRA	0.40	0.14	1196.66	0.20	—

ROME: best single-edit profile (highest PG, near-perfect ESR, CS = 1.00 confirming $P(\text{new}) > P(\text{old})$ in every edit). **MEMIT**: best for batch/sequential use (perfect ESR, lowest LL, low IR, CS = 1.00). **LoRA**: unsuitable for surgical editing across all metrics; CS = 0.20 reveals the model prefers the new target in only 20% of edits despite 4 epochs of fine-tuning.

7. Discussion

Generalization–Locality Trade-off. ROME’s concentrated single-layer update produces a strong directional change that generalizes to paraphrased prompts (PG=0.73) but can leak into unrelated representations (LL=1.50, peaking at 51.77 for one rare subject). MEMIT’s distributed approach across layers 8–12 largely eliminates these locality outliers (LL=1.00) and maintains low sequential interference (IR=0.04), at the cost of generalization breadth (PG=0.56). This trade-off is a principled consequence of update geometry: concentration produces strong generalization; distribution produces strong locality.

Why LoRA Fails at Surgical Editing. LoRA’s optimization objective — minimize cross-entropy over all training examples — is inherently global. Each gradient step simultaneously reshapes representations for all token contexts passing through the adapted layers, with no mechanism to confine changes to a specific subject’s factual memory pathway. The result is catastrophic locality leakage (LL = 1196.66, with some facts reaching LL > 7000), even when training fewer than 0.05% of parameters. This confirms that *parameter efficiency is not the same as edit precision*.

Scale of Evaluation. Our experiments on 100 CounterFact edits provide substantially more statistical power than studies on 5–10 edits. The ROME ESR outliers (2 failures out of 100) and the LoRA locality distribution reveal patterns invisible in small samples.

Covariance Approximation. The original ROME formulation uses a corpus-estimated covariance matrix $C = \mathbb{E}[\mathbf{k}\mathbf{k}^\top]$ over $\sim 1,000$ diversity-sampled sentences. Our ROME implementation uses $C = I$, which suffices for our experiments but may yield sub-optimal key projections at larger scale. Our MEMIT implementation uses the proper regularised covariance ($C = K K^\top + \lambda I$), which is a methodological improvement over the simpler hook-based MEMIT baseline.

Limitations. (1) While 100 edits is an improvement, the full CounterFact benchmark contains 21,919 edits; large-scale experiments are needed. (2) 4-bit quantization may affect editing precision vs. full-precision models. (3) Long-term edit stability across extended usage is not measured. (4) The IR boundary for large sequential batches ($N > 100$) is unknown.

8. Future Work and Conclusion

Conclusion. Surgical knowledge swapping in Qwen2.5-1.5B-Instruct is feasible and effective with closed-form editing methods. Causal tracing identified Layer 10 as the critical factual memory layer (213.0% recovery). ROME delivers near-perfect edit success (0.98) with strong paraphrase generalization for single-fact edits. MEMIT achieves perfect ESR with low inter-edit interference (IR=0.04) and superior locality for batch/sequential use. LoRA fails surgically despite its parameter efficiency, confirming that gradient-based fine-tuning is fundamentally

incompatible with surgical precision.

Future Work.. (1) **Proper covariance estimation:** compute C from 1,000 diverse sentences to improve ROME key projection at scale. (2) **Full-scale evaluation:** run all three methods on the complete 21,919-edit CounterFact benchmark. (3) **Hybrid approach:** combine ROME’s generalization ($PG=0.73$) with MEMIT’s locality ($LL=1.00$). (4) **Multi-hop evaluation:** test whether edits propagate correctly through reasoning chains. (5) **Scaling:** evaluate on 7B and 13B models to assess layer-index predictability.

9. Division of Labor

Rhesha Vinod (Tasks 1 & 2): Cross-platform model loading (CUDA/MPS/CPU), 4-bit NF4 quantization setup via bitsandbytes, CounterFact data pipeline (100 edits), multi-layer activation patching harness, and Layer 10 identification.

Prathiksha Nijamkari (Tasks 1 & 2): Reproducibility infrastructure (seed fixing, JSONL logging, determinism verification), initial layer recovery benchmarks.

Abhishek Soundalgekar (Task 3): Full ROME pipeline: key extraction, 25-step Adam value optimization, rank-one ΔW computation, and forward-hook injection for 4-bit quantized weights. All 100 single-edit experiments with per-edit model restoration.

Aryan Sawant (Task 4): MEMIT multi-layer update system (layers 8–12, scale=0.2 per layer), covariance-based formulation ($C = KK^\top + \lambda I$, CPU-solved), Sherman–Morrison single-fact updates, batch (100 facts simultaneously) and sequential edit experiments, interference rate ($IR=0.04$) analysis.

Siddhanth Salian (Task 5): LoRA baseline (419,840 trainable parameters), 4-epoch AdamW fine-tuning with loss curve, five-metric evaluation framework (ESR, PG, LL, CS, IR) with per-method CS computed immediately post-edit for each method, all comparative visualizations.

All members contributed to project planning, report writing, and presentation preparation.

References

- [1] K. Meng, D. Bau, A. Andonian, and Y. Belinkov, “Locating and editing factual associations in GPT,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, pp. 17359–17372, 2022.
- [2] K. Meng, A. S. Sharma, A. Andonian, Y. Belinkov, and D. Bau, “Mass-editing memory in a transformer,”

in *International Conference on Learning Representations (ICLR)*, 2023.

- [3] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-rank adaptation of large language models,” in *ICLR*, 2022.
- [4] Qwen Team, “Qwen2.5 technical report,” *arXiv:2412.15115*, 2025.
- [5] M. Geva, R. Schuster, J. Berant, and O. Levy, “Transformer feed-forward layers are key-value memories,” in *EMNLP*, pp. 9797–9808, 2021.
- [6] D. Dai, L. Dong, Y. Hao, Z. Sui, B. Chang, and F. Wei, “Knowledge neurons in pretrained transformers,” in *ACL*, pp. 8493–8502, 2022.
- [7] E. Mitchell, C. Lin, A. Bosselut, C. Finn, and C. D. Manning, “Fast model editing at scale,” in *ICLR*, 2022.
- [8] N. De Cao, W. Aziz, and I. Titov, “Editing factual knowledge in language models,” in *EMNLP*, pp. 6491–6506, 2021.
- [9] T. Brown *et al.*, “Language models are few-shot learners,” in *NeurIPS*, vol. 33, pp. 1877–1901, 2020.
- [10] A. Sinitsin, V. Plokhotnyuk, D. Pyrber, S. Popov, and A. Babenko, “Editable neural networks,” in *ICLR*, 2020.